New Strategies to Treat Prescription Opioid Abuse

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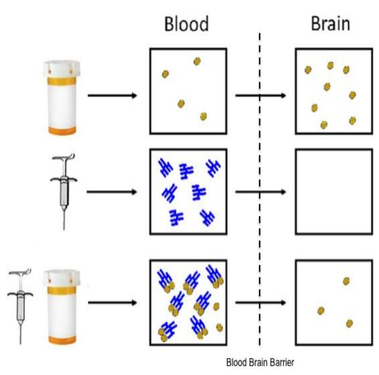
Abstract

Opiate addiction is a social and economic problem worldwide; however, over the past two decades prescription opioid abuse drastically increased in the USA (Compton and Volkow, 2006 [1], Compton and Volkow, 2006 [2]). Treatment for opioid abuse is limited to a few approved medications which exhibit limited clinical efficacy.

Recently, vaccines against abused drugs have shown evidence of clinical efficacy. An oxycodone vaccine has shown preclinical efficacy as a potential treatment targeting oxycodone and hydrocodone, the most commonly abused prescription opioids (Pravetoni et al., 2012).

The goal of this study was to test if the opioid antagonist naltrexone attenuates oxycodone and hydrocodone behavioral effects and if naltrexone may be combined with vaccination against oxycodone. In this study, we found that naltrexone dose-dependently blocked the behavioral effects of oxycodone and hydrocodone in a test of thermal nociception in rats and that naltrexone combined with the oxycodone vaccine increased its efficacy. Vaccines may be used in combination with addiction medications to increase overall clinical efficacy.

Background



Opioid Abuse in USA

- 13 million abuse opioids (including heroin and prescription opioids)

Addiction Vaccine as Alternative Approach

- Long acting and selective
- Potentially less expensive
- Elicit antibodies binding drug, not the receptor
- May be combined with other approved therapies for enhanced efficacy

Study Purpose

- Study if naltrexone (NLTX) blocks the behavioral effects of oxycodone (OXY) and hydrocodone (HYDROC) in a test of thermal nociception (hot plate) in rats (Experiment 1 & 2).
- Study if the OXY-KLH vaccine blocks the behavioral effects of OXY and HYDROC in the hot plate test in rats (Experiment 3).
- Study if combining NLTX with the OXY-KLH vaccine increases efficacy of treatment on OXY behavioral effects (Experiment 4).

References

Compton W.M.; Volkow N.D. Abuse of prescription drugs and the risk of addiction. Drug and Alcohol Dependence 2006;83(Suppl. 1):S4-7

Compton W.M.; Volkow N.D. Major increases in opioid analgesic abuse in the United States: concerns and strategies. Drug and Alcohol Dependence 2006;8(2):103-7.

Pravetoni M, Le Naour M, Harmon TM, Tucker AM, Portoghese PS, Pentel PR. An oxycodone conjugate vaccine elicits drug-specific antibodies that reduce oxycodone distribution to brain and hot-plate analgesia. J Pharmacol Exp Ther. 2012 Apr;341(1):225-32. Epub 2012 Jan 18.

Experiment 1 & 2 Methods

Rats were administered saline or NLTX (0.001-1.0 mg/kg) subcutaneously (s.c.). Fifteen minutes after the saline or NTLX injection, all rats received 2.25 mg/kg OXY (Experiment 1) or 6.75 mg/kg HYDROC (Experiment 2), s.c. (see diagram below). Dose-response curves were created in drug naïve rats. Nociception was measured as hind paw lick, jump or reaching the 60 second cutoff to avoid tissue damage. The maximal possible effect (MPE%) was calculated as: $\left(\frac{postdrug\ latency - baseline\ latency}{60 - baseline\ latency}\right) \times 60$



Experiment 1 & 2 Results

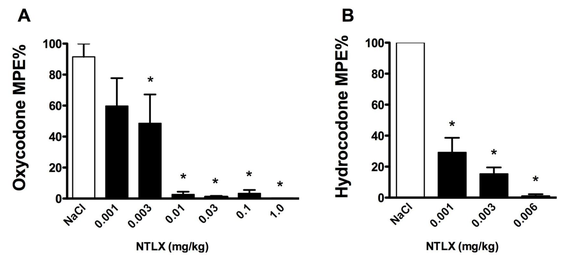
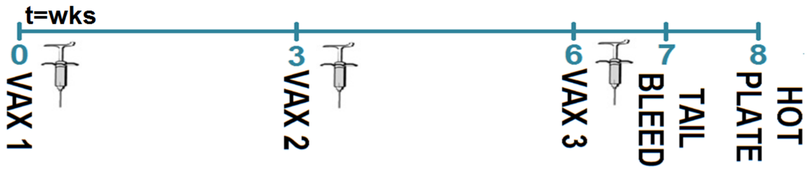


Figure 1: Naltrexone reduces oxycodone and hydrocodone analgesia in the hot plate test.
(A) Dose response of NLTX in drug naïve rats followed 15 minutes later by 2.25 mg/kg OXY, s.c. (B) Dose response of NLTX in naïve rats followed 15 minutes later by 6.75 mg/kg HYDROC, s.c.
NOTE: Statistical symbols: *p<0.05 from saline control.

Experiment 3 & 4 Methods

Rats were immunized with 25 µg of the OXY-KLH vaccine or the control protein KLH (see diagram below) as previously described (Pravetoni et al., 2012). On week 7, OXY-specific serum antibodies were measured by enzyme-linked immunosorbent assay (ELISA) to determine antibody concentrations. On week 8, rats were tested on the hot plate. In Experiment 3, the effect of vaccination was tested on OXY and HYDROC analgesia. In Experiment 4, the effect of NLTX on OXY was tested on rats vaccinated with OXY-KLH or KLH controls.



Experiment 3 Results

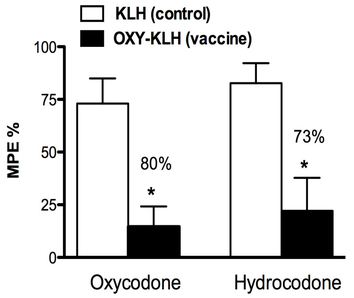


Figure 2: Vaccination with OXY-KLH reduces oxycodone and hydrocodone analgesia in the hot plate test.
Rats immunized with OXY-KLH or KLH protein were tested with 2.25 mg/kg OXY or 6.75 mg/kg HYDROC, s.c.
NOTE: Statistical symbol: *p<0.05 from KLH control.

Experiment 4 Results

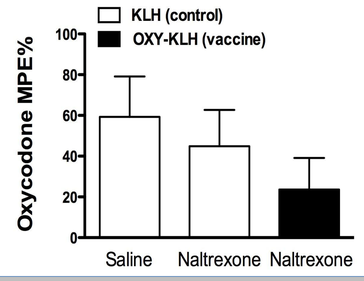


Figure 3: Combining naltrexone with vaccination shows increased effects of treatment on oxycodone analgesia in the hot plate test.
Rats immunized with OXY-KLH or KLH protein were pretreated with saline or NLTX (0.003 mg/kg, s.c.) and tested with 2.25 mg/kg OXY, s.c.. NOTE: no statistical difference was detected.

Conclusion

- Naltrexone blocks the behavioral effects of both oxycodone and hydrocodone in a test of thermal nociception in rats.
- The OXY-KLH vaccine blocks the behavioral effects of both oxycodone and hydrocodone in a test of thermal nociception in rats.
- Combining naltrexone and vaccination enhances the efficacy of naltrexone treatment on oxycodone analgesia.
- Further studies will explore the use of clinically-approved naltrexone formulations in combination with vaccination.

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